12.2 Surface Areas of Prisms and Cylinders

Name _____ Period ____ Date _____

Geometry 12.2 (30-minute plan) · Student sheet

I can…

- find the surface area of a right prism.
- find the surface area of a right cylinder.
- find a lateral area or a missing dimension from a surface area.

Vocabulary · fill in the missing word

Lateral faces — the faces of a _____ that are not bases

Lateral area — the sum of the _____ of the lateral faces (or the curved surface of a cylinder)

Surface area — the sum of the _____ of all the faces or surfaces of a solid

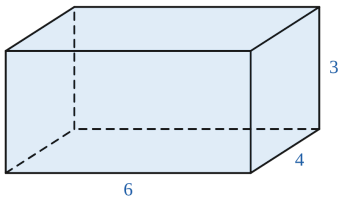
Net — a _____ pattern that forms a solid when it is folded

Part 1 · Notes and examples (fill in each note, then work the example)

Surface area of a right prism: $S = 2B +$ _____, where B is the area of a base, P is the perimeter of a base, and h is the height.

Example 1

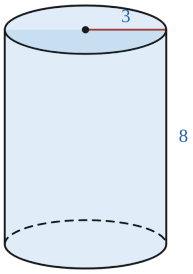
Find the surface area of the rectangular prism.



Surface area of a right cylinder: $S = 2\pi r^2 +$ _____

Example 2

Find the surface area of the cylinder. Round to the nearest hundredth.



Lateral area: leave out the _____: $L = Ph$ for a prism, $L = 2\pi rh$ for a cylinder.

Example 3

A soup can has a radius of 1.5 inches and a height of 4.5 inches. How much paper does its label (the curved side only) need? Round to the nearest hundredth.

Part 2 · Partner practice

Partner practice

A cylinder has a diameter of 10 and a height of 6. Find its surface area. Round to the nearest hundredth.